**Pavel Aleksandrov**
**Position:** Unity / C# Developer (Gameplay & Netcode Generalist)

- **Location:** Georgia (Open to Remote / Relocation)
- **Telegram:** @Alindel163
- **Email:** alindel222@gmail.com
- **GitHub / Portfolio:** alindel163.github.io/porfolio/
- **Languages:** Russian (Native), English (B2 — comfortable with technical documentation and interviews)

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## SUMMARY

A dedicated Unity / C# Developer with a strong software engineering foundation in algorithms, network communication, and hardware-software interaction. Experienced in developing and deploying commercial WebGL/HTML5 games with a focus on cross-platform performance. Highly versatile and hands-on across the entire production pipeline, equally comfortable writing robust netcode, optimizing 3D assets (Blender/AccuRig), or implementing UI layouts. Bringing over 3 years of collaborative team-leading experience in technical education environments.

**Career Goal:** Seeking a long-term, full-time opportunity within a game development studio. I approach game creation with an engineering mindset, prioritizing stability, clean architecture (KISS/SOLID), and project deadlines. Highly adaptable and ready to contribute wherever the team needs support—whether it's multiplayer sync, optimization, or UI implementation. My goal is to become a reliable pillar for the team and grow alongside the studio's long-term projects.

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## HARD SKILLS

- **Engine & Core:** Unity (3D/2D), C#, C++, SQL, Git (Git-flow).
- **Networking & Backend:** Netcode for GameObjects, .NET 8/9, ASP.NET Core Web API, WebSockets (TCP), Docker, Redis, PostgreSQL.
- **Tech Art & Optimization:** Blender (modeling & mesh optimization), ActorCore AccuRig / Mixamo (skeletal rigging), WebGL Optimization (Object Pooling, Draw Calls reduction, texture atlases).
- **Gameplay & Architecture:** UI Toolkit, New Input System, Hierarchical State Machines (HFSM), Pathfinding (A*), Async Pipelines (System.Threading.Channels).
- **R&D & Tools:** Prototyping workflows utilizing AI asset pipelines (Trellis 2, Fooocus, SAM 2).

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## WORK EXPERIENCE

**Independent Contract Developer / Unity Game Developer** | *Freelance & Consulting*
*2025 — Present (1 year)*

- **Commercial WebGL Delivery:** Developed and launched commercial HTML5/WebGL indie games for Yandex Games and PlayMarket platforms.
- **Performance Profiling:** Conducted deep WebGL optimizations (Object Pooling, texture management, rendering budget allocation) to ensure a stable 60 FPS on low-spec mobile units.
- **Collaborative Development:** Partnered with distributed mini-teams on a contract basis, participating in code reviews and facilitating the successful delivery of 20+ diverse Unity/C# projects.

**Engineering Team Lead** | *University College*
*2023 — 2025 (2 years)*

- **Team Collaboration:** Facilitated development cycles for 10+ cross-functional student teams working on interactive 3D simulations and game prototypes. Focused on clear task breakdown and fostering a healthy team environment.
- **Architecture & Quality Support:** Guided system design choices, established constructive peer code review practices, and helped resolve technical roadblocks without personal ego.
- **Technical Training:** Developed and delivered a practical, deep-dive C++ course (memory safety, pointers, custom data structures, debugging tools) to enhance the team's engineering core.

**Lead Software Engineer (R&D)** | *iSchool Engineering Network*
*2017 — 2023 (6 years)*

- **Ecosystem Design:** Designed and scaled a reliable software ecosystem used for hardware automation and simulation platforms.
- **Low-Level Software:** Engineered C++ backend modules and asynchronous data synchronization protocols for over 30 interactive hardware deployment schemes.

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**TECHNICAL PROJECTS (R&D / PORTFOLIO)**

- **Co-op Boss Raid Prototype (C# / Unity / Netcode for GameObjects)**
  A multiplayer action prototype featuring NetworkVariable state synchronization, RPC calls, and network object spawning. Implemented client-side position interpolation for smooth puppet rendering.
- **Advanced Combat AI System (C# / Unity)**
  An extensible NPC behavior architecture built on Hierarchical State Machines (HFSM) managing complex state transitions (chase, tactical retreat, combat), paired with a custom target-locking system.
- **R&D Case: High-Speed 3D Prototyping Pipeline (Trellis 2 / Blender / AccuRig / Unity)**
  Researched an accelerated asset integration pipeline, reducing the workflow from raw concept to an animated 3D unit down to 40 minutes. Stack: Fooocus -> SAM 2 -> Trellis 2 -> Blender (mesh cleanup) -> AccuRig (auto-rigging). Features interactive web-based visualization.
- **Custom Runtime Interpreter inside Unity (C#)**
  Developed a lexical/syntax parser and runtime interpreter for a C++ subset inside Unity. Built to deeply understand and experiment with low-level engine memory management concepts at the C# layer.

- **Multiplayer TCP Relay Server (Pure C# / .NET Core / WebSockets)**
  A lightweight, framework-free network relay server handling multi-threaded state synchronization for a shared virtual simulation in real time.

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